计算系统生物学作业2

宋思逸

2400010816

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1. Competitive and uncompetitive inhibition

a.

$$[E]_{tot} = [E] + [ES] + [EI] \tag{1}$$

For competitive inhibitors,

$$K_I \triangleq \frac{[E][I]}{[EI]} \tag{2}$$

Now that we assume that $I_0 \gg [E]_{tot}$, we have $[I] \approx I_0 (I_0 = [I] + [EI])$

Hence

$$[EI] = \frac{[E]I_0}{K_1} \tag{3}$$

Hence

$$[E]_{tot} = [E] + [ES] + \frac{[E]I_0}{K_I} = [E](1 + \frac{I_0}{K_I}) + [ES] \tag{4}$$

Now we try to deduce the adjusted Michaelis-Menten equation.

$$K_M = \frac{[E][S]}{[ES]} \implies [E] = \frac{K_M[ES]}{S_0}(S_0 \gg [E]_{tot} \implies [S] \approx S_0) \tag{5}$$

Based on 1, we have

$$[E]_{tot} = [E] + [ES] + \frac{[E]I_0}{K_I} = [E]\left(1 + \frac{I_0}{K_I}\right) + [ES] \tag{6}$$

Based on 5, we have

$$[E]_{tot} = \left(\frac{K_M[ES]}{S_0}\right)\left(1 + \frac{I_0}{K_I}\right) + [ES] \tag{7}$$

Denote $1 + \frac{I_0}{K_I} \triangleq \alpha$

$$[E]_{tot} = [ES]\left(\frac{\alpha K_M}{S_0} + 1\right) \implies [E]_{tot} = [ES]\left(\frac{\alpha K_M + S_0}{S_0}\right) \tag{8}$$

Hence

$$[ES] = \frac{[E]_{tot}S_0}{\alpha K_M + S_0} \tag{9}$$

Because

$$V_0 = k_{cat}[ES] \tag{10}$$

and

$$V_{max} = k_{cat}[E]_{tot} \tag{11}$$

We have

$$V_0 = \frac{V_{max}S_0}{\alpha K_M + S_0} \tag{12}$$

Maximal turnover rate is scarcely affected as $[S](S_0) \to +\infty$

$$\lim_{S_0 \to +\infty} \frac{V_{max}S_0}{\alpha K_M + S_0} = V_{max} \tag{13}$$

For half the maximum rate, we have $S_0 = \alpha K_M > K_M$, so we will need higher substrate concentration.

b.

$$K_I' \triangleq \frac{[ES][I]}{[ESI]} \tag{14}$$

Similarly, we assume $[I] = I_0$

$$[ESI] = \frac{[ES]I_0}{K_I'} \tag{15}$$

$$[E]_{tot} = [E] + [ES] + [ESI] \tag{16}$$

hence

$$[E]_{tot} = [E] + [ES]\left(1 + \frac{I_0}{K_I'}\right) \tag{17}$$

Similarly, we denote $1 + \frac{I_0}{K_I'} \triangleq \beta$

Maximal turnover rate decreases (to $\frac{V_{max}}{\beta}$), and the substrate concentration at half the maximum rate decreases (to $\frac{K_M}{\beta}$). This can be deduced from

$$V_0 = \frac{V_{max}S_0}{K_M + \beta S_0} = \frac{\frac{V_{max}}{\beta}S_0}{\frac{K_M}{\beta} + S_0} \tag{18}$$

c. Assume that the system is at detailed balance, we can define the following dissociation constants.

$$K_m = \frac{[E][S]}{[ES]}, K_I = \frac{[E][I]}{[EI]}, K_I' = \frac{[ES][I]}{[ESI]} \tag{19}$$

Since

$$[E]_{tot} = [E] + [ES] + [EI] + [ESI] \tag{20}$$

We have

$$[E]_{tot} = [E] + [ES] + \frac{[E][I]}{K_I} + \frac{[ES][I]}{K_I'} = [E]\left(1 + \frac{[I]}{K_I}\right) + [ES]\left(1 + \frac{[I]}{K_I'}\right) \tag{21}$$

Denote $\alpha \triangleq 1 + \frac{[I]}{K_I}, \beta \triangleq 1 + \frac{[I]}{K_I'}$

We have

$$[E]_{tot} = \alpha[E] + \beta[ES] \tag{22}$$

Now that $[E] = \frac{K_m[ES]}{[S]}$, we have

$$[ES] = \frac{[E]_{tot}[S]}{\alpha K_m + \beta[S]} \tag{23}$$

Notice that

$$v = k_{cat}[ES] \tag{24}$$

and

$$V_{max} = k_{cat}[E]_{tot} \tag{25}$$

We now have

$$v = \frac{k_{cat}[E]_{tot}[S]}{\alpha K_m + \beta[S]} = \frac{V_{max}[S]}{\alpha K_m + \beta[S]} \tag{26}$$

i.e. the modified equation

$$v = \frac{\frac{V_{max}}{\beta}[S]}{\frac{\alpha}{\beta}K_m + [S]} \tag{27}$$

d. Based on the graph, we have a decreased V_{max}, and K_m stays the same (approximately). Hence we consider it a mixed inhibitor.

2. Stability analysis of dynamical systems

a. (1)

$$\frac{dx}{dt} = rx - x^2 \tag{28}$$

Set $\frac{dx}{dt} = 0$, we get two fixed points $x^* = 0$ and $x^* = r$.

Since $f(x) = -\frac{dV}{dt}$, we have

$$V(x) = -\int f(x)\,dx = -\frac{r}{2}x^2 + \frac{1}{3}x^3 \tag{29}$$

If $r > 0$:

$$f(x) \begin{cases} > 0, & 0 < x < r \\ < 0, & x > r \quad \text{or} \quad x < 0 \end{cases} \tag{30}$$

Hence $x^* = 0$ is unstable, while $x^* = r$ is stable.

For $V(x)$, $x = 0$ is a local maximum point, while $x = r$ is a local minimum point.

If $r = 0$: $f(x) \leq 0$, $x^* = 0$ is a half-stable point. For $V(x)$, $x^* = 0$ is an inflection point.

If $r < 0$: Similarly we hae $x^* = 0$ is stable, while $x = r^*$ is unstable. For $V(x)$, $x^* = r$ is a local maximum point, and $x^* = 0$ is a local minimum point.

(2)

$$\frac{dx}{dt} = rx - x^3 \tag{31}$$

Set $\frac{dx}{dt} = 0$, we get $x(r - x^2) = 0$. $V(x) = -\int (rx - x^3)\,dx = -\frac{r}{2}x^2 + \frac{1}{4}x^4$

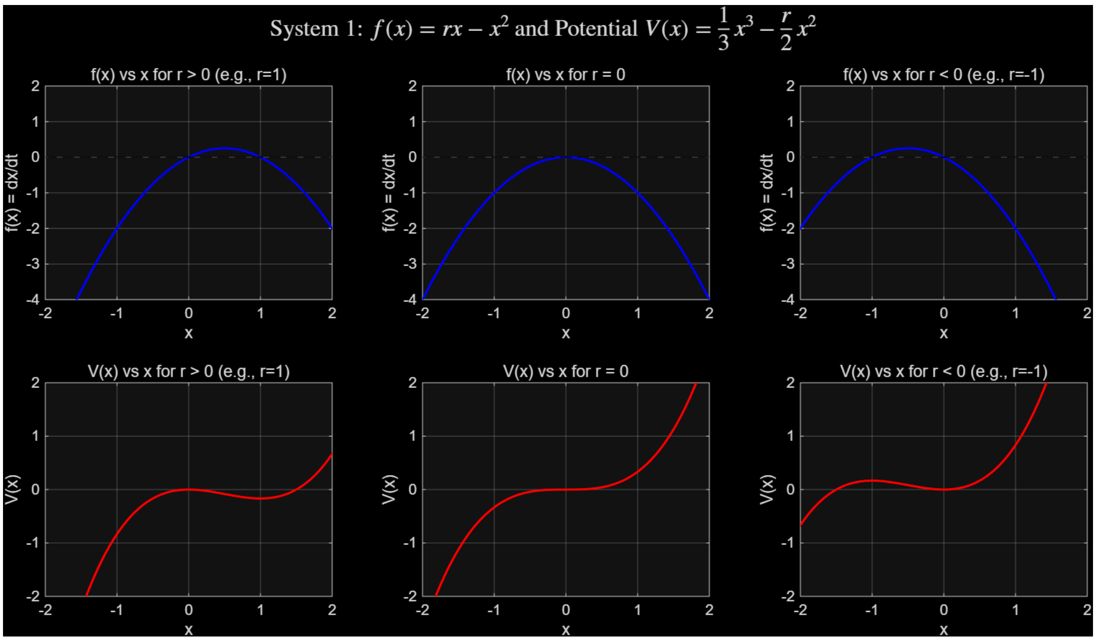
Now we discuss different situations.

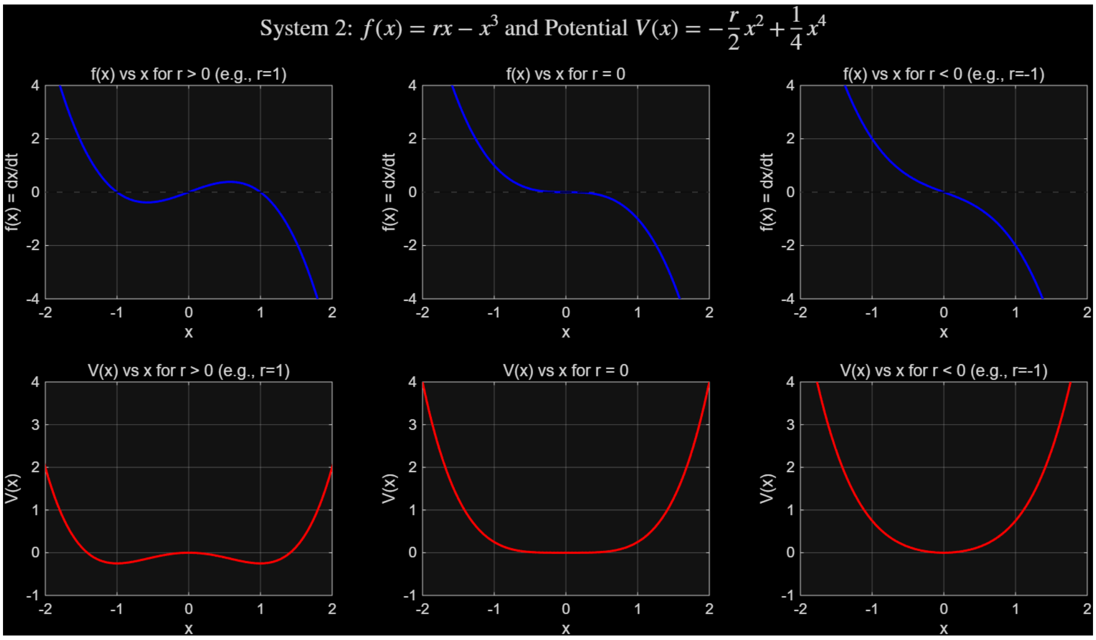
If $r > 0$: the fixed points are $x^* = 0, x^* = \pm\sqrt{r}$. $x^* = \pm\sqrt{r}$ are stable, while $x^* = 0$ is unstable.

If $r = 0$, $x^* = 0$ is stable.

If $r < 0$, $x^* = 0$ is stable.

Now we give the graphs of the systems above.





b. (1)

$$\frac{dx}{dt} = x(3 - x - y), \frac{dy}{dt} = y(2 - x - y) \tag{32}$$

Set $\frac{dx}{dt} = \frac{dy}{dt} = 0$, we have $x = 0; x+y = 3$ and $y = 0; x+y = 2$, we get fixed points:$(0, 0), (0, 2), (3, 0)$
If we denote $f(x, y) \triangleq x(3 - x - y), g(x, y) \triangleq y(2 - x - y)$, the Jacobi matrix

$$J(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} = \begin{pmatrix} 3 - 2x - y & -x \\ -y & 2 - x - 2y \end{pmatrix} \tag{33}$$

$$J(0, 0) = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \tag{34}$$

Eigenvalues are $\lambda_1 = 3, \lambda_2 = 2$, $(0, 0)$ is unstable.

$$J(0, 2) = \begin{pmatrix} 1 & 0 \\ -2 & -2 \end{pmatrix} \tag{35}$$

Eigenvalues are $\lambda_1 = 1, \lambda_2 = -2$, $(0, 2)$ is saddle.

$$J(3, 0) = \begin{pmatrix} -3 & -3 \\ 0 & -1 \end{pmatrix} \tag{36}$$

Eigenvalues are $\lambda_1 = -3, \lambda_2 = -1$, $(3, 0)$ is stable node.
Nullclines:
x-nullcline: $x = 0$ and $x + y = 3$
y-nullcline: $y = 0$ and $x + y = 2$
Basin of attraction: Since $(3, 0)$ is the only stable point, $x > 0, y \geq 0$ is its basin of attraction.
(2)

$$\frac{dx}{dt} = x(3 - 2x - y), \frac{dy}{dt} = y(2 - x - y) \tag{37}$$

Set $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$, we get $x = 0; 2x + y = 3$ and $y = 0; x + y = 2$. We get fixed points $(0,0), (0,2), (1.5,0), (1,1)$

Similarly, by directly calculating eigenvalues, or by judging its relationship with 0 by calculating trace and determinant of the Jacobi matrix, we have unstable node $(0,0)$, saddle $(0,2), (1.5,0)$, and stable node $(1,1)$.

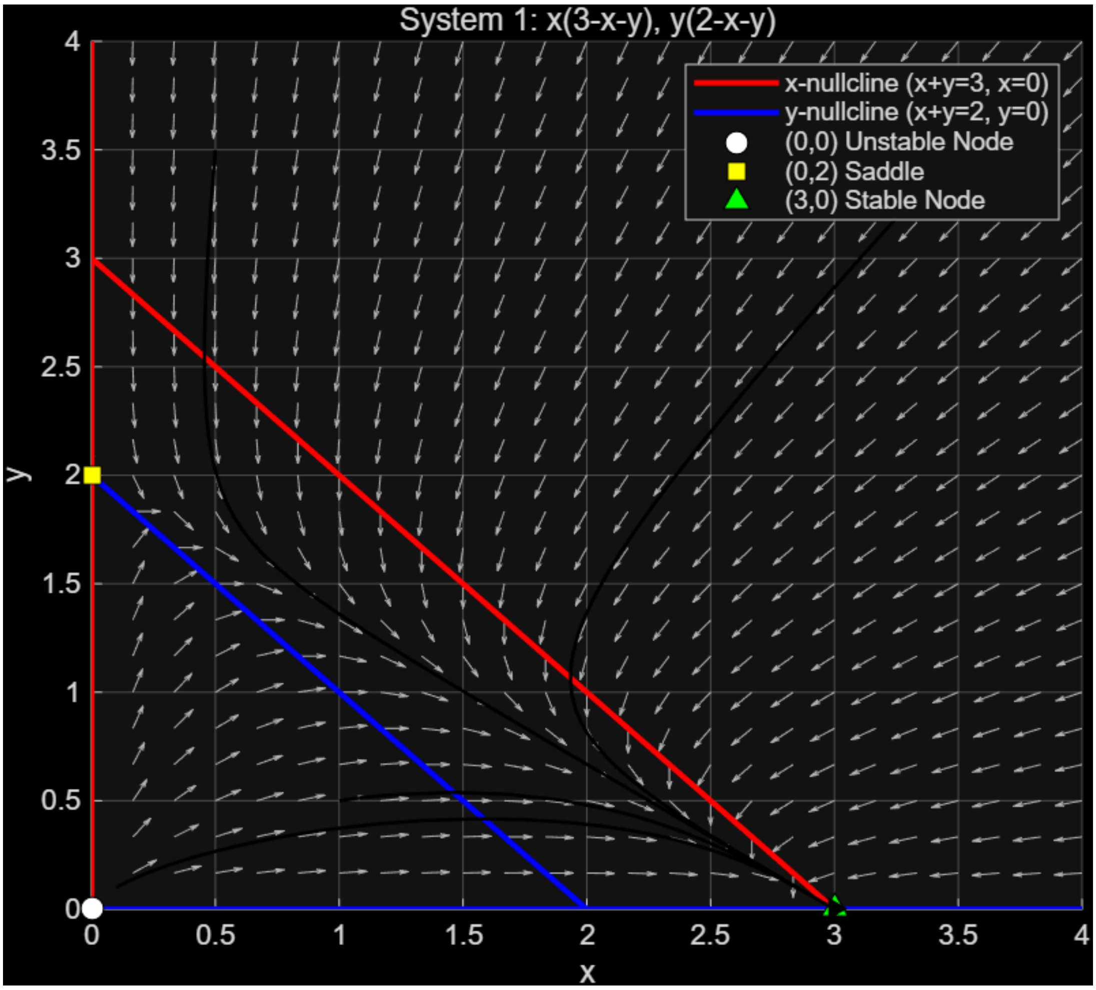
Nullclines:

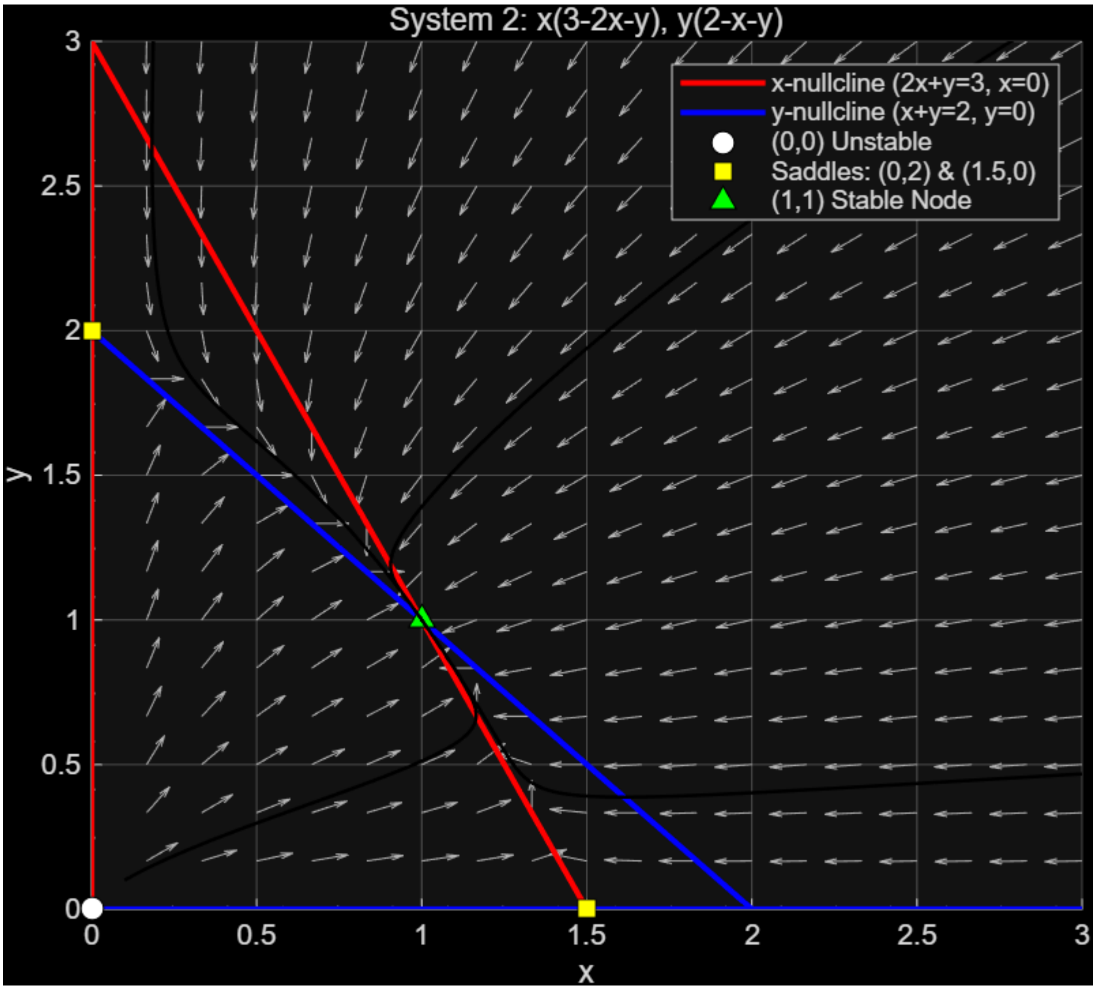
x-nullcline:$x = 0$ and $2x + y = 3$

y-nullcline:$y = 0$ and $x + y = 2$

Basin of attraction: $x > 0, y > 0$ for $(1,1)$

Now we give the graphs of the two systems.





3. Positive feedback and bistability

a. (1) The strongly cooperative binding implies the promoter is in two states: D and DX_AX_B. Notice that

$$[DX_AX_B] = K_1K_2[D][X]^2 \tag{38}$$

and

$$[D]_{tot} = [D] + [DX_AX_B] \tag{39}$$

Hence we have the portion:

$$P_D = \frac{[D]}{[D] + [DX_AX_B]} = \frac{1}{1 + K_1K_2[X]^2} \tag{40}$$

$$P_{DX_AX_B} = \frac{[DX_AX_B]}{[D] + [DX_AX_B]} = \frac{K_1K_2[X]^2}{1 + K_1K_2[X]^2} \tag{41}$$

Denote $[X] \triangleq x$, production rate can be expressed as

$$f(x) = \nu_0 P_D + \nu_1 P_{DX_AX_B} = \frac{\nu_0}{1 + K_1K_2x^2} + \frac{\nu_1K_1K_2x^2}{1 + K_1K_2x^2} = \frac{\nu_0 + \nu_1K_1K_2x^2}{1 + K_1K_2x^2} \tag{42}$$

Consider dilution(notice that degradation can be neglected), we have

$$f(x) = \frac{\nu_0 + \nu_1K_1K_2x^2}{1 + K_1K_2x^2} - \gamma x \tag{43}$$

(2) Steady state solutions occur when

$$\frac{\nu_0 + \nu_1 K_1 K_2 x^2}{1 + K_1 K_2 x^2} = \gamma x \tag{44}$$

$f(x)$ and $g(x)$ can intersect at 1,2 or 3 points.

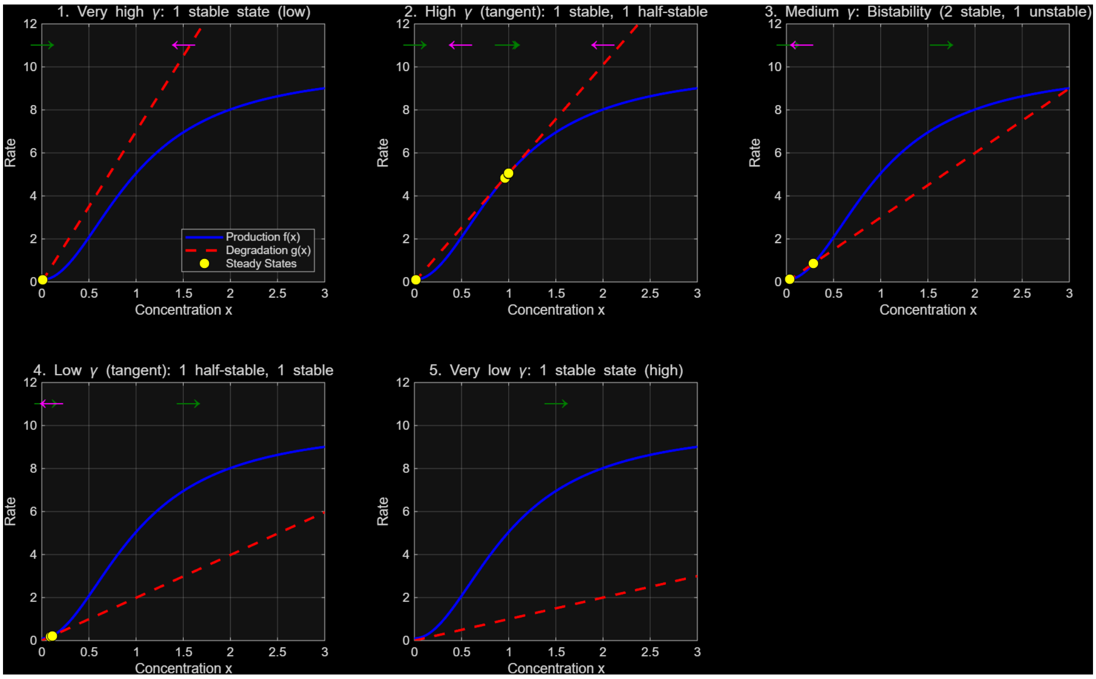
1) one stable point

2) 2 steady state point, one is stable and another is half-stable

3) 3 steady state, 1 and 3 are stable, while 2 is unstable.

4) 2 steady state, one is half-stable, and another is stable.

5) 1 stable point



b.

The model is given by

$$\frac{du}{dt} = \frac{\alpha}{1+v^2} - u \tag{45}$$

$$\frac{dv}{dt} = \frac{\alpha}{1+u^2} - v \tag{46}$$

Steady state can be calculated by setting $\frac{du}{dt} = 0$, $\frac{dv}{dt} = 0$, i.e.

$$u = \frac{\alpha}{1+v^2} \qquad v = \frac{\alpha}{1+u^2} \tag{47}$$

According to the symmetry of the equation, we have a solution $u = v$, and we get

$$u^3 + u - \alpha = 0 \tag{48}$$

$f(u) = u^3 + u - \alpha$ is monotone increasing on $\mathbb{R}$, so the equation has only one solution u_0.

The Jacobi matrix is defined as

$$J = \begin{pmatrix} \frac{\partial \dot{u}}{\partial u} & \frac{\partial \dot{u}}{\partial v} \\ \frac{\partial \dot{v}}{\partial u} & \frac{\partial \dot{v}}{\partial v} \end{pmatrix} \tag{49}$$

$$\frac{\partial \dot{u}}{\partial u} = -1 \tag{50}$$

$$\frac{\partial \dot{u}}{\partial v} = -\frac{2\alpha v}{(1+v^2)^2} \tag{51}$$

$$\frac{\partial \dot{v}}{\partial u} = -\frac{2\alpha u}{(1+u^2)^2} \tag{52}$$

$$\frac{\partial \dot{v}}{\partial v} = -1 \tag{53}$$

Hence at (u_0, u_0)

$$J_0 = \begin{pmatrix} -1 & -\frac{2\alpha u_0}{(1+u_0^2)^2} \\ -\frac{2\alpha u_0}{(1+u_0^2)^2} & -1 \end{pmatrix} \tag{54}$$

Denote $A = \frac{2\alpha u_0}{(1+u_0^2)^2}$

$$J_0 = \begin{pmatrix} -1 & -A \\ -A & -1 \end{pmatrix} \tag{55}$$

$$\det(J_0 - \lambda I) = \begin{vmatrix} -1-\lambda & -A \\ -A & -1-\lambda \end{vmatrix} = (-1-\lambda)^2 - A^2 = 0 \tag{56}$$

Hence

$$(-1-\lambda) = \pm A \implies \lambda_1 = -1 + A, \quad \lambda_2 = -1 - A \tag{57}$$

To make (u_0, u_0) unstable, since $\lambda_2 = -1 - A < 0 (A > 0)$, we have $\lambda_1 = A - 1 > 0$ i.e.

$$A > 1 \implies \frac{2\alpha u_0}{(1+u_0^2)^2} > 1 \tag{58}$$

At steady state,

$$u_0 = \frac{\alpha}{1+u_0^2} \tag{59}$$

$$A = \frac{2[u_0(1+u_0^2)]u_0}{(1+u_0^2)^2} = \frac{2u_0^2}{1+u_0^2} \tag{60}$$

Now that $A > 1$

$$\frac{2u_0^2}{1+u_0^2} > 1 \implies 2u_0^2 > 1 + u_0^2 \implies u_0^2 > 1 \implies u_0 > 1 \implies \alpha > 2 \tag{61}$$

In conclusion, $\boldsymbol{\alpha > 2}$ give the system bistability.

4. Kinetic Proofreading

a. Assume that the reaction is at steady-state.

For the correct complex Cc

$$\frac{d[Cc]}{dt} = k'_C[C][c] - k_C[Cc] - W[Cc] = 0 \implies [Cc] = \frac{k'_C[C][c]}{k_C + W} \tag{62}$$

For the wrong complex Dc:

$$\frac{d[Dc]}{dt} = k'_D[D][c] - k_D[Dc] - W[Dc] = 0 \implies [Dc] = \frac{k'_D[D][c]}{k_D + W} \tag{63}$$

The definition goes:

$$f \triangleq \frac{v_{\text{error}}}{v_{\text{correct}}} = \frac{W[Dc]}{W[Cc]} = \frac{[Dc]}{[Cc]} \tag{64}$$

Hence

$$f = \frac{\frac{k'_D[D][c]}{k_D+W}}{\frac{k'_C[C][c]}{k_C+W}} \tag{65}$$

Given $[C] = [D]$, we have

$$f = \frac{k'_D(k_C + W)}{k'_C(k_D + W)} \tag{66}$$

For the minimum value f_0, we assume $W \ll k_C$. This implies

$$f_0 = \frac{k'_D k_C}{k'_C k_D} = \frac{K_D}{K_C} \tag{67}$$

b. The steady state equation is modified to

$$\frac{d[Xc]}{dt} = k'_X[X][c] + m_X[Xc^*] - (k_X + m'_X)[Xc] = 0 \tag{68}$$

$$\frac{d[Xc]}{dt} = m'_X[Xc] + l'_X[X][c] - (m_X + l_X + W)[Xc^*] = 0 \tag{69}$$

where $X = C, D$.

Hence

$$[Xc] = \frac{k'_X[X][c] + m_X[Xc^*]}{k_X + m'_X} \tag{70}$$

$$m'_X\left(\frac{k'_X[X][c] + m_X[Xc^*]}{k_X + m'_X}\right) + l'_X[X][c] - (m_X + l_X + W)[Xc^*] = 0 \tag{71}$$

$$[Xc^*]\left[(m_X + l_X + W) - \frac{m'_X m_X}{k_X + m'_X}\right] = [X][c]\left[\frac{m'_X k'_X}{k_X + m'_X} + l'_X\right] \tag{72}$$

We get

$$[Xc^*] = [X][c]\frac{m'_X k'_X + l'_X(k_X + m'_X)}{k_X m_X + l_X(k_X + m'_X) + W(k_X + m'_X)} \tag{73}$$

The error rate is given by

$$f = \frac{v_D}{v_C} = \frac{W[Dc^*]}{W[Cc^*]} = \frac{[Dc^*]}{[Cc^*]} \tag{74}$$

i.e.

$$f = \frac{m'_D k'_D + l'_D(k_D + m'_D)}{m'_C k'_C + l'_C(k_C + m'_C)} \cdot \frac{k_C m_C + l_C(k_C + m'_C) + W(k_C + m'_C)}{k_D m_D + l_D(k_D + m'_D) + W(k_D + m'_D)} \tag{75}$$

The restriction of detailed balance is

$$k'_X \cdot m'_X \cdot l_X = k_X \cdot m_X \cdot l'_X \tag{76}$$

$$\frac{m'_X}{m_X} = \frac{l'_X k_X}{l_X k'_X} \tag{77}$$

Therefore, we have

$$[Xc^*] = [X][c]\frac{\frac{l'_X}{l_X}\Big(m_X k_X + l_X k_X + l_X m'_X\Big)}{\Big(m_X k_X + l_X k_X + l_X m'_X\Big) + W(k_X + m'_X)} \tag{78}$$

i.e.

$$[Xc^*] = [X][c]\frac{\frac{l'_X}{l_X}}{1 + W \cdot \tau_X} \tag{79}$$

where

$$\tau_X = \frac{k_X + m'_X}{m_X k_X + l_X(k_X + m'_X)} = \frac{1}{l_X + \frac{m_X k_X}{k_X + m'_X}} \tag{80}$$

The error rate

$$f = \frac{[Dc^*]}{[Cc^*]} = \left(\frac{l'_D/l_D}{l'_C/l_C}\right) \cdot \frac{1 + W\tau_C}{1 + W\tau_D} \tag{81}$$

Notice that $l_D > l_C$ and $k_D > k_C$, we have $\tau_D < \tau_C$. Since $W > 0$, we have $\frac{1+W\tau_C}{1+W\tau_D} \geq 1$. Notice that $f_0 = \frac{l'_D/l_D}{l'_C/l_C}$, we have come to the conclusion

$$f = f_0\left(\frac{1 + W\tau_C}{1 + W\tau_D}\right) \geq f_0 \tag{82}$$

c. Since $m = 0$ and $l'_c = 0$, we have

$$\mathrm{X} + \mathrm{c} \xrightleftharpoons[\text{k''X}]{\text{k'X}} \mathrm{Xc} \xrightarrow{m'} Xc^* \xrightarrow{W} \text{product} \tag{83}$$

Also

$$Xc^* \xrightarrow{l_X} X + c \tag{84}$$

$m' \ll k_C$ implies quasi-equilibrium.

$$k'_X[X][c] \approx k_X[Xc] \implies [Xc] = \frac{k'_X}{k_X}[X][c] \tag{85}$$

$$\frac{d[Xc^*]}{dt} = m'[Xc] - l_X[Xc^*] - W[Xc^*] = 0 \tag{86}$$

$$m'[Xc] = (l_X + W)[Xc^*] \tag{87}$$

Hence

$$[Xc^*] = \frac{m'}{l_X + W}[Xc] = \frac{m'}{l_X + W}\left(\frac{k'_X}{k_X}\right)[X][c] \tag{88}$$

Calculate the error rate f:

$$f = \frac{W[Dc^*]}{W[Cc^*]} = \frac{[Dc^*]}{[Cc^*]} \tag{89}$$

$$f = \frac{\frac{m'}{l_D+W}\left(\frac{k'_D}{k_D}\right)[D][c]}{\frac{m'}{l_C+W}\left(\frac{k'_C}{k_C}\right)[C][c]} = \left(\frac{k'_D k_C}{k'_C k_D}\right) \cdot \left(\frac{l_C + W}{l_D + W}\right) \tag{90}$$

$$f = f_0 \cdot \left(\frac{l_C + W}{l_D + W}\right) \tag{91}$$

Assume $W \ll l_C$ and $W \ll l_D$

$$\frac{l_C + W}{l_D + W} \approx \frac{l_C}{l_D} \tag{92}$$

$$f \approx f_0 \cdot \frac{l_C}{l_D} \tag{93}$$

$$\frac{l_D}{l_C} = \frac{k_D}{k_C} \implies \frac{l_C}{l_D} = \frac{k_C}{k_D} \tag{94}$$

$$f_0 = \frac{k'_D k_C}{k'_C k_D} \approx \frac{k_C}{k_D} \tag{95}$$

$$f \approx f_0 \cdot \left(\frac{l_C}{l_D}\right) = f_0 \cdot f_0 = f_0^2 \tag{96}$$

5.Circadian Clocks

a. (A)

$$\frac{dx}{dt} = v_x + k_x \frac{A_1}{A_1 + y} - \gamma_x x \tag{97}$$

$$\frac{dy}{dt} = v_y + k_y \frac{x}{A_2 + x} - \gamma_y y \tag{98}$$

(B)

$$\frac{dx}{dt} = v_x + k_x \frac{x^2}{(A_3)^2 + x^2} \frac{A_1}{A_1 + y} - \gamma_x x \tag{99}$$

$$\frac{dy}{dt} = v_y + k_y \frac{x}{A_2 + x} - \gamma_y y \tag{100}$$

Basal transcription rate (Without activator or inhibitor): v_x, v_y.

Maximal transcription rate: k_x, k_y.

Degradation rate: γ_x, γ_y.

b.

Now we give explanation of the fractions.

$$\frac{A_1}{A_1 + y} \tag{101}$$

This fraction stands for the portion of X promoter that stays active, without combining with Y inhibitor.

$$\frac{x}{A_2 + x} \tag{102}$$

This fraction stands for the portion of Y promoter that stays active since it is successfully combined with activator X.

$$\frac{x^2}{A_3^2 + x^2} \tag{103}$$

This fraction stands for the portion of X promoter that is activated by the cooperative binding of two molecules of X, and hence be auto-activated.

c. Given $A_2 \gg x$, we have

$$\frac{dy}{dt} = v_y + k_y \frac{x}{A_2 + x} - \gamma_y y \approx v_y + k_y \frac{x}{A_2} - \gamma_y y \tag{104}$$

Now we define some parameters:

$$\bar{y} = \frac{y}{A_1} \quad \Rightarrow \quad y = A_1 \bar{y} \tag{105}$$

$$\bar{x} = \frac{x}{x_0} \quad \Rightarrow \quad x = x_0 \bar{x} \tag{106}$$

$$\bar{t} = \gamma_y t \quad \Rightarrow \quad t = \frac{\bar{t}}{\gamma_y} \tag{107}$$

$$\frac{d(A_1\bar{y})}{d(\bar{t}/\gamma_y)} = v_y + \frac{k_y}{A_2}(x_0\bar{x}) - \gamma_y(A_1\bar{y}) \tag{108}$$

$\Longrightarrow$

$$\frac{d\bar{y}}{d\bar{t}} = \frac{v_y}{\gamma_y A_1} + \frac{k_y x_0}{\gamma_y A_1 A_2}\bar{x} - \bar{y} \tag{109}$$

Denote

$$\bar{v}_y = \frac{v_y}{\gamma_y A_1}, \bar{k}_y = \frac{k_y x_0}{\gamma_y A_1 A_2} \tag{110}$$

We have

$$\frac{d\bar{y}}{d\bar{t}} = \bar{v}_y + \bar{k}_y\bar{x} - \bar{y} \tag{111}$$

Now we calculate $\frac{d\bar{x}}{d\bar{t}}$ for system (B).

$$\gamma_y A_3\frac{d\bar{x}}{d\bar{t}} = v_x + k_x\frac{(A_3\bar{x})^2}{(A_3)^2 + (A_3\bar{x})^2}\frac{A_1}{A_1 + A_1\bar{y}} - \gamma_x(A_3\bar{x}) \tag{112}$$

$$\gamma_y A_3\frac{d\bar{x}}{d\bar{t}} = v_x + k_x\frac{\bar{x}^2}{1+\bar{x}^2}\frac{1}{1+\bar{y}} - \gamma_x A_3\bar{x} \tag{113}$$

$$\frac{d\bar{x}}{d\bar{t}} = \frac{v_x}{\gamma_y A_3} + \frac{k_x}{\gamma_y A_3}\frac{\bar{x}^2}{1+\bar{x}^2}\frac{1}{1+\bar{y}} - \frac{\gamma_x}{\gamma_y}\bar{x} \tag{114}$$

$$\frac{d\bar{x}}{d\bar{t}} = \frac{\gamma_x}{\gamma_y}\left(\frac{v_x}{\gamma_x A_3} + \frac{k_x}{\gamma_x A_3}\frac{\bar{x}^2}{1+\bar{x}^2}\frac{1}{1+\bar{y}} - \bar{x}\right) \tag{115}$$

Denote

$$\bar{\gamma}_x = \frac{\gamma_x}{\gamma_y}, \bar{v}_x = \frac{v_x}{\gamma_x A_3}, \bar{k}_x = \frac{k_x}{\gamma_x A_3} \tag{116}$$

We have

$$\frac{d\bar{x}}{d\bar{t}} = \bar{\gamma}_x\left(\bar{v}_x + \bar{k}_x\frac{\bar{x}^2}{1+\bar{x}^2}\frac{1}{1+\bar{y}} - \bar{x}\right) \tag{117}$$

Similarly for system (A), we have

$$\frac{d\bar{x}}{d\bar{t}} = \bar{\gamma}_x(\bar{v}_x + \bar{k}_x\frac{1}{1+\bar{y}} - \bar{x}) \tag{118}$$

d. For system (A)

x-nullcline: $y = \frac{4}{x-0.1} - 1$

y-nullcline: $y = 2x$

Fixed point:

$$\begin{cases} y = \frac{4}{x-0.1} - 1 \\ y = 2x \end{cases} \tag{119}$$

$$2x^2 + 0.8x - 4.1 = 0 \tag{120}$$

$x_0 \approx 1.245, y_0 = 2x_0 \approx 2.49$, and the fixed point $(1.245, 2.49)$.

The Jacobi matrix

$$J_{(1.245,2.49)} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}\Bigg|_{(1.245,2.49)} = \begin{pmatrix} -10 & -3.28 \\ 2 & -1 \end{pmatrix} \tag{121}$$

$tr(A) = -11 < 0, \det(A) = 16.56 > 0 \Longrightarrow$ the fixed point of A is stable.

For system (B)

x-nullcline: $x - 0.1 = \frac{4x^2}{(1+x^2)(1+y)}$

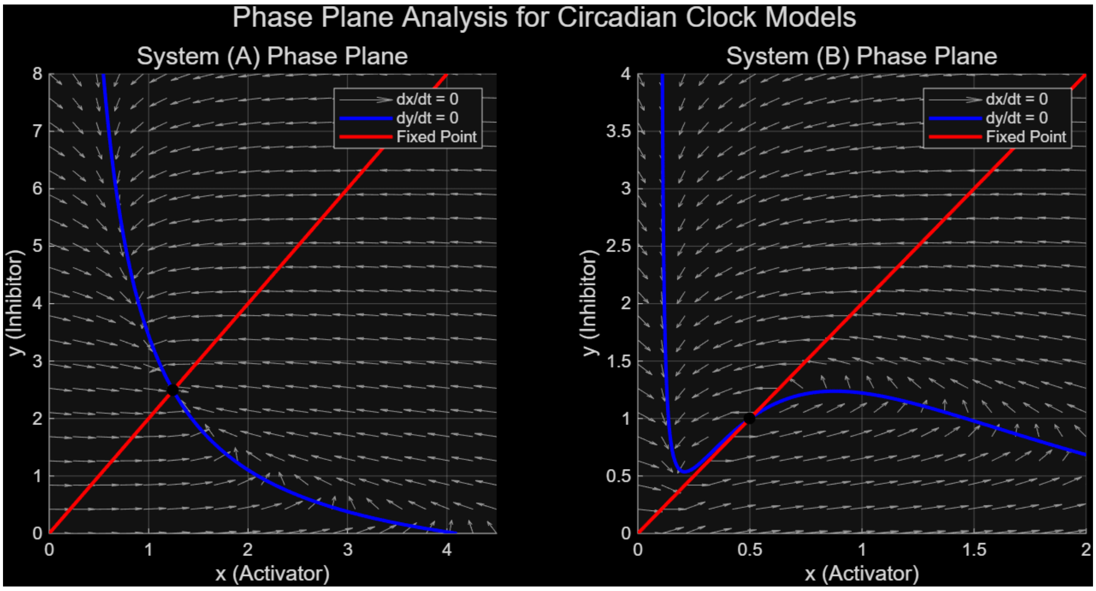
y-nullcline: $y = 2x$

We get the fixed point $(0.5, 1.0)$.

The Jacobi matrix

$$J_{(0.5,1.0)} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}\Bigg|_{(0.5,1.0)} = \begin{pmatrix} 2.8 & -2.0 \\ 2 & -1 \end{pmatrix} \tag{122}$$

$tr(A) = 1.8 > 0, \det(A) = 1.2 > 0 \implies$ the fixed point of B is unstable.



e.

证明. The Jacobi matrix:

$$J = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} = \begin{pmatrix} -\gamma_x & -\frac{\gamma_x k_x}{(1+y)^2} \\ k_y & -1 \end{pmatrix} \tag{123}$$

$tr(J) = -\gamma_x - 1 < 0, \det(J) = \gamma_x + \frac{\gamma_x k_x k_y}{(1+y)^2} > 0$

Hence we have only one fixed point and it is stable. The system will always converge to a stable fixed point. □

f. 1 Find the fixed point

$$\frac{dx}{dt} = \gamma_x \left(v_x + k_x \frac{x^2}{1+x^2} \frac{1}{1+y} - x \right) = 0 \tag{124}$$

$$\frac{dy}{dt} = v_y + k_y x - y = 0 \tag{125}$$

We know that $v_x = 0.1, v_y = 0.0, k_x = 4.0, k_y = 2.0$. Notice we already know that $\frac{dy}{dt} = 0 \implies y = 2x$, hence $x = 0.5, y = 1.0$, and the fixed point $(0.5, 1.0)$.

The Jacobi matrix

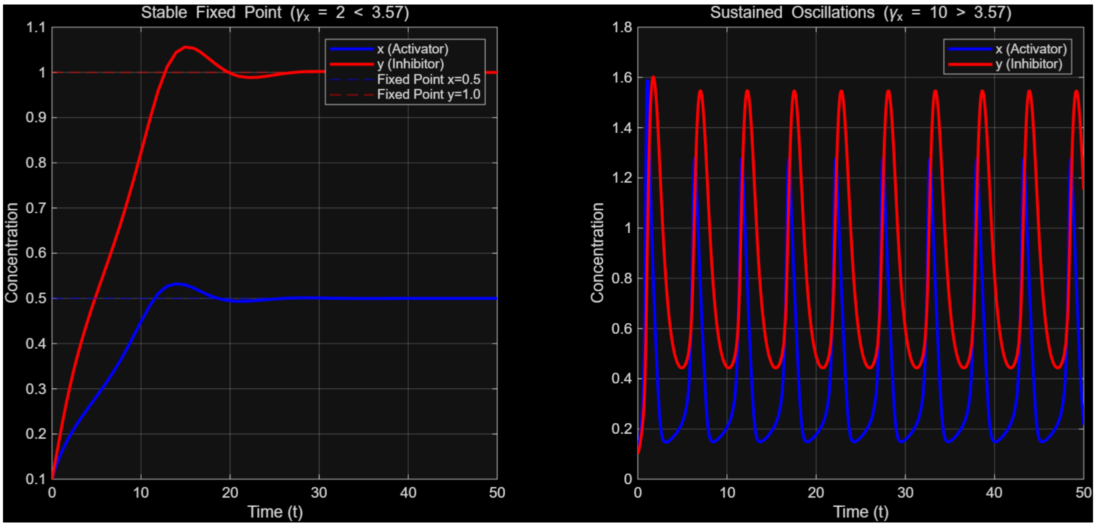
$$J = \begin{pmatrix} \gamma_x \left[\frac{k_x}{1+y} \frac{d}{dx} \left(\frac{x^2}{1+x^2} \right) - 1 \right] & \gamma_x \left[k_x \frac{x^2}{1+x^2} \frac{-1}{(1+y^2)} \right] \\ k_y & -1 \end{pmatrix} = \begin{pmatrix} 0.28\gamma_x & -0.2\gamma_x \\ 2 & -1 \end{pmatrix} \tag{126}$$

Hence we have $tr(J) = 0.28\gamma_x - 1, \det(J) = 0.12\gamma_x$.

Apparently, $\gamma_x > 0$, hence $\det(J) > 0$. So the fixed point is unstable $\iff$ $tr(J) > 0$, i.e., $\gamma_x > \frac{25}{7}$, and this leads to oscillations.

A large γ_x implies that X promoter must be much faster than Y inhibitor, which is a typical relaxation oscillation. To be more specific, X accumulates quickly at first. Y responds slowly, and will have to wait for adequate concentration before inhibiting X. This process goes on periodically.

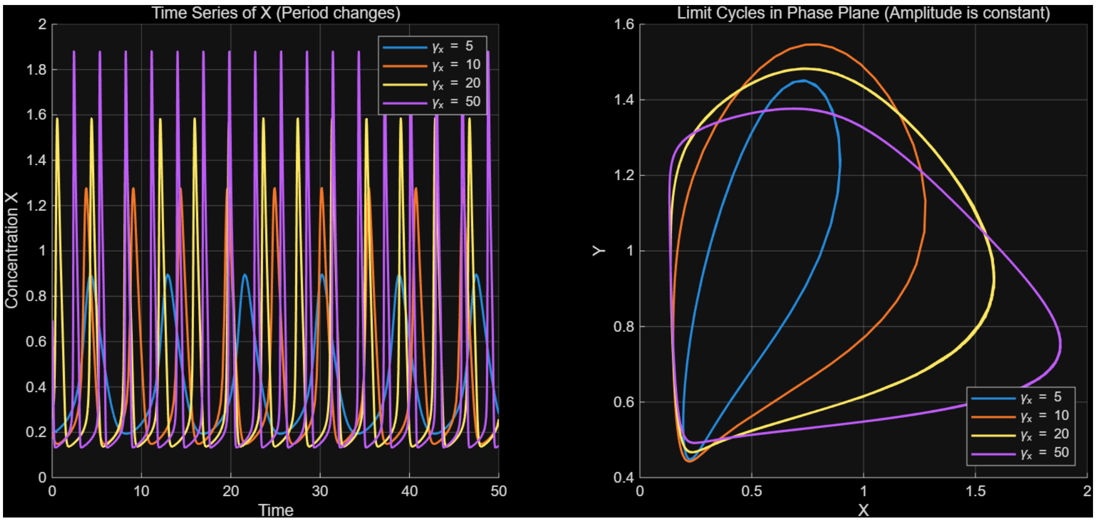
2. According to the critical value we have just calculated, we choose $\gamma_x = 2.0$ and $\gamma_x = 10.0$.



3. As γ_x is near the critical value, the amplitude changes violently. As it gets large enough, the amplitude goes to a certain limit.

For a repressilator, trying to adjust the period may lead to violent change of the amplitude.

As for tunability, system B seems to be great.



4.

An example is the mammalian circadian clock. In the core clock network, the transcription factor complex CLOCK:BMAL1 activates expression of PER and CRY, while accumulated PER/CRY complexes inhibit CLOCK:BMAL1 activity, forming a delayed negative-feedback loop capable of oscillation. This corresponds closely to motif (B), where an activator promotes an inhibitor, and the inhibitor suppresses the activator.

Reference:Reppert, S., Weaver, D. Coordination of circadian timing in mammals. Nature 418, 935–941 (2002). https://doi.org/10.1038/nature00965